Asset Pricing: Fama-MacBeth

Yifan Qi

University of Edinburgh Business School, Edinburgh, UK

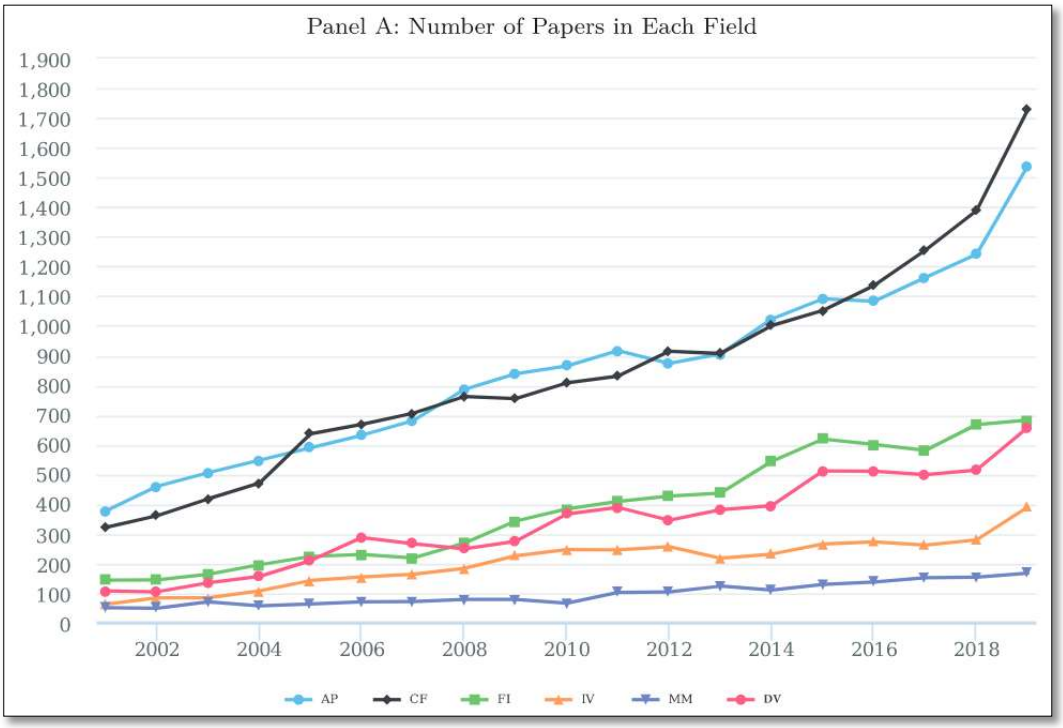
Y.Q-18@sms.ed.ac.uk

https://www.linkedin.com/in/yifan-qi-88100117a

Keywords: Asset Pricing, Two-Step Regression Approach, Fama-MacBeth

1 What is Asset Pricing?

(a) Asset Pricing



Word cloud and publication number charts by Rui Dai (2023)

Asset Pricing Theory: helps us understand the fundamental relationship between risk and return in financial markets. Investors demand compensation for bearing risk.

Asset Pricing Model: Capital Asset Pricing Model (CAPM), Arbitrage Pricing Theory (APT), and multi-factor models provide frameworks to quantify this relationship systematically.

They forms the foundation of rational investment decision-making (see Appendix I), e.g., risk-return tradeoff principle. Practically, it guides portfolio construction and optimization, risk management practices, corporate financing decisions, and the valuation of complex financial instruments.

The Foundation of Rational Investment Decision-Making

The foundation of rational investment decision-making is the risk-return tradeoff principle (Markowitz, 1952). This principle states that investors should be compensated for taking additional risk with the expectation of higher returns. In a rational framework, investors make decisions by:

1. Evaluating expected returns of different assets
2. Assessing the risks associated with those returns
3. Making optimal allocation decisions based on their risk preferences

This foundation is formalized in asset pricing models like the Capital Asset Pricing Model (CAPM), which establishes a linear relationship between systematic risk (beta) and expected returns. The underlying assumption is that investors are utility-maximizing and risk-averse[1], meaning they require additional compensation for bearing non-diversifiable risk (systematic risk).

For instance, corporations use asset pricing models to determine appropriate discount rates for project evaluation, while asset managers apply these theories to develop investment strategies (portfolio management) that deliver superior risk-adjusted returns (positive alpha[2] - the investment is generating returns beyond what would be expected

[1] **Risk aversion**: it describes the behavioral preference of investors to avoid uncertainty when possible. A risk-averse investor, when faced with two investments with the same expected return but different risks, will choose the less risky option.

[2] **This positive alpha** can result from: Market inefficiencies that the investor has successfully exploited, the manager's skill in security selection or market timing, or exposure to risk factors not captured by the market

given its level of systematic risk).

E.g. Capital Asset Pricing Model (CAPM) theory:

$$E(ri) = r_f + \beta_i[E(r_m) - r_f]$$

Where:

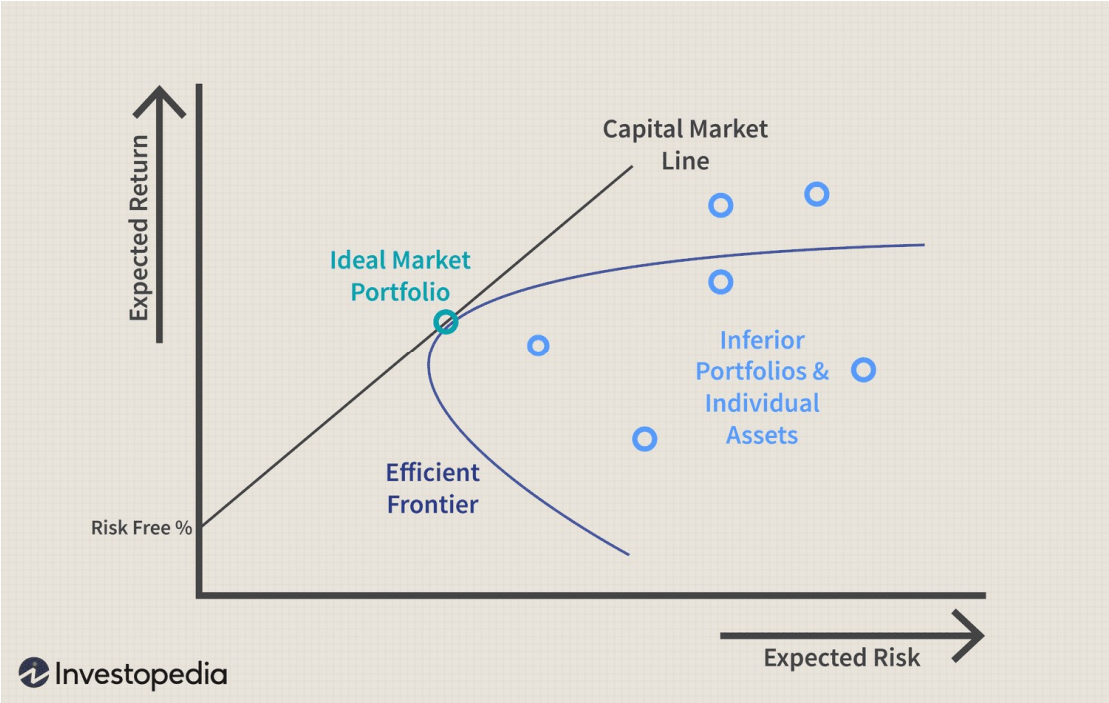
$E(ri)$ is the expected return of asset i

r_f is the risk-free rate

β_i is the beta of asset i (measure of systematic risk)

$E(r_m)$ is the expected market return

$[E(r_m) - r_f]$ is the market risk premium[3]



Efficient Frontier: The blue curved line represents all optimal portfolios that provide the highest expected return for a given level of risk, or the lowest risk for a given level of expected return. Each point on this curve represents an efficient portfolio.

Capital Market Line: The straight line extending from the Risk-Free Rate point and

---

beta alone.

[3] **Risk premium**: the additional return investors expect to earn for holding riskier assets compared to risk-free assets.

tangent to the Efficient Frontier. This line represents the optimal risk-return combinations available to investors when they can borrow and lend at the risk-free rate by combining the risk-free asset with the market portfolio, i.e., Separation Theorem (Tobin, 1985).

Ideal Market Portfolio: This is the point where the Capital Market Line touches the Efficient Frontier. It represents the theoretically optimal portfolio of risky assets, typically interpreted as the market portfolio. According to CAPM theory, all investors should hold this portfolio in varying proportions combined with the risk-free asset to match their risk preferences.

Inferior Portfolios & Individual Assets: The blue circles located inside the Efficient Frontier represent individual assets or portfolios that are suboptimal. These choices are inferior because investors can find portfolios on the Efficient Frontier that offer either higher returns for the same risk or lower risk for the same return.

Expected Risk vs. Expected Return Trade-off: The horizontal axis represents Expected Risk (typically measured by standard deviation or variance), while the vertical axis represents Expected Return. The graph demonstrates the relationship between risk and return, and how combining different assets can optimize this trade-off.

Asset Pricing Research: illuminates market efficiency and anomalies. While the Efficient Market Hypothesis suggests that prices fully reflect all available information, research has uncovered numerous "anomalies" such as the size effect, value premium. These findings challenge our understanding of market functioning.

It contributes significantly to financial stability and regulatory policy. Understanding how asset prices form and fluctuate helps predict and mitigate financial crises, providing central banks and regulatory bodies with theoretical foundations for their policy decisions. E.g. the 2008 financial crisis and Intermediary Asset Pricing.

2 Testing the CAPM with Fama-MacBeth Regression

Methodology

The Fama-MacBeth (Fama and MacBeth, 1973) regression is a two-stage regression methodology widely used in empirical asset pricing to test factor models and estimate risk premiums. It allows researchers to empirically test whether this two-parameter structure adequately explains cross-sectional variation in asset returns. The procedure

examines if beta is the only relevant risk factor that determines expected returns, as predicted by CAPM.

Rationale of introducing it:

1) Testing Factor Pricing Ability (Pricing Effectiveness)

The fundamental function of Fama-MacBeth tests is to evaluate whether risk factors significantly explain cross-sectional variations in asset returns.

2) Broad Academic Recognition and Conventional Practice

Classic asset pricing research literature employs Fama-MacBeth tests as the standard methodology, establishing a convention in asset pricing research. Reviewers and readers widely recognize and accept the Fama-MacBeth testing procedure, making it a standardized analytical method in the field.

Assumptions:

"Two-parameter" portfolio model: the framework where securities' expected returns are determined by just two parameters:

1) The risk-free rate
2) A risk premium associated with systematic risk (beta)

Market equilibrium: It refers to the assumption that markets have reached a state where all investors have optimized their portfolios given their preferences, and prices have adjusted to clear markets[4]. In equilibrium, only systematic risk should be priced because diversifiable risk can be eliminated without sacrificing expected return. Only under the equilibrium framework can the price of asset returns have predictable and explainable laws to follow. When the market price deviates from the equilibrium state, investors will carry out arbitrage behavior, thereby driving the market back to equilibrium.

Efficient Set Theorem (Markowitz, 1952)**: The market portfolio is actually efficient (lies on the efficient frontier) -** investors will only choose portfolios from the efficient frontier—portfolios that maximize expected return for a given level of risk or minimize risk for a given level of expected return.

---

[4] Clear markets: supply and demand for all assets are balanced, which means that at the current price, all willing sellers have found buyers, and vice versa.

Assumes perfect capital markets with no transaction or information costs: this assumption simplifies the theoretical model underlying the Fama-MacBeth tests and has several important implications:

1. Frictionless Trading: All investors can trade without costs, meaning that arbitrage opportunities would be immediately exploited, pushing prices to their equilibrium levels.
2. Information Efficiency: All relevant information is freely available and immediately incorporated into prices, supporting the notion that only systematic risk factors should be priced.
3. Test Implications: Any observed deviations from the model predictions (like non-linear relationships between risk and return, or significant alphas) could indicate either: market inefficiencies, transaction costs that prevent perfect arbitrage, information asymmetries. missing risk factors not captured by beta.It establishes a clear theoretical benchmark against which actual market behaviour can be compared.

In practice, these assumptions rarely hold perfectly, which is partly why the empirical tests often find deviations from the CAPM predictions. The Fama-MacBeth procedure helps quantify these deviations and has been instrumental in the development of more sophisticated asset pricing models that account for additional risk factors beyond beta.

Hypothesises

C1: Linearity — Expected return on a security is a linear function of its risk.

C2: Risk Measure—β as a complete measure of risk. $\gamma_{2t} = \gamma_{3t} = \eta_{pt} = 0$

C3: Positive Return-Risk Tradeoff—Higher risk should be associated with higher expected return. $\gamma_{1t} > 0$

S-L: Sharpe-Lintner Hypothesis—the intercept represents risk-free. $\gamma_{0t} - R_{ft} = 0$

Theoretical Model:

$$R_{pt} = \gamma_{0t} + \gamma_{1t}\hat{\beta}_{p,t-1} + \gamma_{2t}\hat{\beta}^2_{p,t-1} + \gamma_{3t}\bar{s}_{p,t-1}(\widehat{\epsilon_i}) + \eta_{pt}$$

R_{pt}: portfolio returns

$\hat{\beta}_{p,t-1}$, $\hat{\beta}^2_{p,t-1}$: estimated portfolio beta and its squares,

$\bar{s}_{p,t-1}(\hat{\epsilon}_i)$: idiosyncratic volume of CAPM residual in estimating beta.

Data: Monthly returns for common stocks over a period of time.

Forming a certain number of portfolios based on ranked beta values

Fama-MacBeth t-ratios:

Cross-sectional regression coefficients for each phase (e.g $\gamma_{j,t}$) Take the average over the time series: $\overline{\gamma_j} = \frac{1}{T}\sum_{t=1}^{T}\gamma_{j,t}$

t-statistics for testing the previous hypothesis that $t(\bar{\hat{\gamma}}_j) = \frac{\bar{\hat{\gamma}}_j}{s(\bar{\hat{\gamma}}_j)/\sqrt{n}}$

where $\bar{y}_j$:Time series mean, standard error $s(\bar{y}_j)$

Two-stage regression approach:

Stage 1: estimates past beta;

Stage 2: testing future returns on beta.

This methodology first conducts separate regressions for each cross-sectional period, then performs t-tests on the time-series means of the coefficients, providing an intuitive and effective verification of each factor's pricing ability.

Example: The base test of the CAPM

The regression model is as follows:

$$r_{i,t} = \lambda_{0,t} + \lambda_{m,t}\hat{\beta}_{im,t-1} + \sigma_{i,t} \qquad i = 1, \dots, N,$$

where $r_{i,t}$ is the industry i excess return at time t, $\hat{\beta}_{im,t-1}$ is the estimate of the beta from regressing US industry I returns on the US market index based on the previous <u>60 months</u> of data, for a period ending at time t−1. Each month, the

Fama-MacBeth regression produces estimates of the intercept, $\lambda_{0,t}$, and the market premium $\lambda_{m,t}$. Use the average coefficients over time and the Fama-MacBeth t-ratios to test the hypothesis that the ***average premium associated with the market risk is positive*** (C3) and that the ***average pricing error is zero*** (C2).

Data source and description: The number of industries is at least 20. The frequency of data is monthly. The number of time-series observations is at least 360. The cross-sectional observations, as well as industry-specific firm attribute should come from DataStream (Thomson Reuters). As a risk-free rate use the one-month US T-bill.

Step	**Type**	**Regression Equation**	**Output**
1	First-pass (Time-series)	$r_{i,t} = \alpha_i + \beta_{i,t} r_{m,t} + \epsilon_{i,t}$ $(r_{m,t}\ can\ be\ other\ common\ factor)$	Estimated betas for each asset
2	Second-pass (Cross-sectional)	$r_{i,t} = \lambda_{0,t} + \lambda_{m,t}\hat{\beta}_{i,t} + \mu_{i,t}$	Risk premium estimate for each period

3	Final step (Time-series averaging)	$\bar{\hat{\lambda}}_j = \frac{1}{T}\sum_{t=1}^{T}\hat{\lambda}_{j,t}$ $t(\bar{\hat{\lambda}}_j) = \frac{\bar{\hat{\lambda}}_j}{s(\bar{\hat{\lambda}}_j)/\sqrt{n}}$	Significance test of risk premium parameter

1. collect data

1) Expected return 2) market return 3) risk free rate

Suppose we have data over 372 months 1990jan-2020dec(31years*12months), 30 industries.

2. Stage 1: generate initial beta ($\hat{\boldsymbol{\beta}}$) by time series test.

Regress $r_{i,t} = \alpha_i + \beta_{i,t} r_{m,t} + \epsilon_{i,t}$ by a 60-month rolling window (e.g. $\hat{\beta}_{1,61} = \beta_{1,1}$, $\hat{\beta}_{1,62} = \beta_{1,2}$)

3. Stage 2: use the $\hat{\boldsymbol{\beta}}$ to regress for generating $\boldsymbol{\lambda_0}$ & $\boldsymbol{\lambda_m}$

Regress $r_{i,t} = \lambda_{0,t} + \lambda_{m,t}\hat{\beta}_{i,t} + \mu_{i,t}$

Generate residuals and Fama-Macbeth t-statistics of λ_0 & λ_m:

Example result:

	λ_0	λ_m
Mean	-0.176	0.165
t	-16.810	0.469

Practice Source:

https://github.com/ZearoGon/Asset-Pricing_Fama-MacBeth.git

Data Source:

https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html

Reference

Fama, E. F., & MacBeth, J. D. (1973). Risk, return, and equilibrium: Empirical tests. *Journal of political economy*, *81*(3), 607-636.

Markowitz, H. (1952). "Portfolio Selection." *The Journal of Finance*, 7(1), 77-91.

Rui Dai, Lawrence Donohue, Qingyi (Freda) Drechsler, Wei Jiang, Dissemination, Publication, and Impact of Finance Research: When Novelty Meets Conventionality, *Review of Finance*, Volume 27, Issue 1, February 2023, Pages 79–141

Tobin, J. (1958). "Liquidity Preference as Behavior Towards Risk." *The Review of Economic Studies*, 25(2), 65-86.

Appendix I

Development Timeline of Rational Investment Decision-Making Theory:

Year	Scholar(s)	Key Contribution	Key Publication	Significance
1952	Harry Markowitz	Modern Portfolio Theory	"Portfolio Selection," *Journal of Finance*	Introduced the mathematical framework for portfolio diversification and established the efficient frontier concept. Demonstrated how to optimize the risk-return tradeoff through diversification.
1958	James Tobin	Separation Theorem	"Liquidity Preference as Behavior Towards Risk," *Review of Economic Studies*	Extended Markowitz's work by introducing the risk-free asset and proving that all investors should hold the same portfolio of risky assets regardless of risk preferences. Led to the Capital Market Line concept.
1961-1966	William Sharpe	Capital Asset Pricing Model (CAPM)	"Capital Asset Prices: A Theory of Market Equilibrium under Conditions of Risk," *Journal of Finance* (1964)	Developed a model that quantifies the relationship between systematic risk and expected return. Introduced beta as a measure of market risk.
1965-1966	John Lintner	Independent development of CAPM	"The Valuation of Risk Assets and the Selection of Risky Investments in Stock Portfolios and Capital	Independently derived the CAPM, adding to its theoretical robustness. Explored implications for capital budgeting.

Year	Scholar(s)	Key Contribution	Key Publication	Significance
			Budgets," *Review of Economics and Statistics* (1965)	
1966	Jan Mossin	Independent development of CAPM	"Equilibrium in a Capital Asset Market," *Econometrica*	Completed the trio of scholars who independently developed CAPM, cementing its place in financial theory.
1970	Eugene Fama	Efficient Market Hypothesis	"Efficient Capital Markets: A Review of Theory and Empirical Work," *Journal of Finance*	Formalized the theory that stock prices reflect all available information, classifying market efficiency into weak, semi-strong, and strong forms.
1973	Fischer Black, Myron Scholes, Robert Merton	Option Pricing Theory	"The Pricing of Options and Corporate Liabilities," *Journal of Political Economy* (Black & Scholes)	Developed a mathematical model for pricing options and derivatives, expanding understanding of risk pricing in financial markets.
1976	Stephen Ross	Arbitrage Pricing Theory (APT)	"The Arbitrage Theory of Capital Asset Pricing," *Journal of Economic Theory*	Proposed a multi-factor model as an alternative to CAPM, suggesting that returns can be predicted using multiple risk factors, not just market risk.
1992	Eugene Fama, Kenneth French	Three-Factor Model	"The Cross-Section of Expected Stock Returns," *Journal of Finance*	Extended CAPM by adding size and value factors to explain stock returns, addressing empirical failures of the

Year	Scholar(s)	Key Contribution	Key Publication	Significance
				single-factor CAPM.
1997	Mark Carhart	Four-Factor Model	"On Persistence in Mutual Fund Performance," *Journal of Finance*	Added momentum as a fourth factor to the Fama-French model, further improving the explanation of stock returns.
2015	Eugene Fama, Kenneth French	Five-Factor Model	"A Five-Factor Asset Pricing Model," *Journal of Financial Economics*	Added profitability and investment factors to their three-factor model, further refining asset pricing theory.